PRACTICAL 4

Aim

To develop and execute a MapReduce program using Hadoop to process NCDC historical weather data and determine the maximum temperature recorded for each year.

Problem Definition

A meteorological research organization maintains historical weather records collected from various weather stations worldwide. The organization uses Hadoop to analyze large climate datasets and study long-term temperature trends.

As a Data Analytics Engineer, the task is to process the National Climatic Data Center (NCDC) weather dataset using Hadoop MapReduce and identify the maximum temperature recorded for each year.

The practical includes uploading the NCDC weather dataset into HDFS, examining the structure of weather records, developing a Mapper to extract the year and temperature, generating intermediate key-value pairs, and developing a Reducer to determine the maximum temperature for each year.

The MapReduce job is then executed on Hadoop, and the output containing the yearly maximum temperatures is stored in HDFS for further analysis.

Objectives

- Understand the working of Hadoop MapReduce. • Understand the structure of NCDC weather data.
- Upload the weather dataset into HDFS.
- Read and parse weather records using the Mapper.
- Extract the **year and temperature** from each record.
- Generate intermediate key-value pairs in the Mapper.
- Process grouped records using the Reducer.
- Find the maximum temperature for each year.
- Execute the MapReduce job on Hadoop.
- Store the output in HDFS.
- Analyze yearly temperature trends.

- Identify the year with the highest recorded temperature.

Commands Used

Command	Purpose
hadoop version	Display installed Hadoop version
jps	Check running Hadoop Java processes
hdfs dfs -mkdir	Create HDFS directories hdfs dfs -put
	Upload dataset to HDFS hdfs dfs -ls List files and
directories in HDFS hdfs dfs -cat	Display HDFS file
contents hdfs dfs -rm	Delete files from HDFS hdfs dfs -rm -r
	Delete directories recursively hadoop jar Execute
the MapReduce JAR file	
hdfs dfs -cat <output>	Display MapReduce output

Procedure

1. Open the Command Prompt, PowerShell, or Linux Terminal.
2. Verify that Java and Hadoop are installed correctly.
3. Start the Hadoop services.
4. Verify the Hadoop processes using jps.
5. Create a directory in HDFS for the NCDC weather dataset.
6. Upload the NCDC weather dataset into HDFS.
7. Examine the structure and format of the weather records.
8. Create the **Mapper program**.
9. Read each weather record in the Mapper.
10. Extract the **year** from the date field.
11. Extract and parse the temperature value from the record.

12. Generate intermediate key-value pairs in the form:

    Key = Year, Value = Temperature

13. Create the **Reducer program**.
14. Receive all temperature values belonging to the same year.
15. Compare the temperature values for each year.
16. Select the maximum temperature for every year.
17. Compile the Mapper and Reducer Java programs.
18. Create a JAR file containing the MapReduce program.
19. Execute the MapReduce job using Hadoop.
20. Store the output in HDFS.
21. Display the generated output using the HDFS command.
22. Analyze the maximum temperature recorded for each year.
23. Identify the year having the highest temperature overall.
24. Stop Hadoop services after completing the practical.

Applications

- Climate data analysis
- Weather forecasting
- Historical temperature analysis
- Global warming studies
- Environmental monitoring
- Meteorological research
- Large-scale climate data processing
- Long-term weather trend analysis
- Big Data analytics
- Scientific data processing

Key Questions / Analysis

Q1. How is weather data parsed and processed using MapReduce?

The NCDC weather dataset is stored as records containing information such as date, temperature, weather station, and other meteorological attributes. The Mapper reads each record, extracts the year and temperature, and generates a key-value pair in the form (year, temperature).

The Reducer receives all temperature values belonging to the same year and compares them to determine the maximum temperature.

Q2. What key-value pairs are generated by the Mapper?

The Mapper generates key-value pairs where:

Key = Year

Value = Temperature

For example:

(1990, 35)

(1990, 38)

(1991, 36)

(1991, 40)

These pairs are then grouped by year before being passed to the Reducer.

Q3. How does the Reducer identify the maximum temperature for a given year?

The Reducer receives all temperature values associated with a particular year. It compares each temperature value and maintains the largest value encountered.

For example:

1990 → [35, 38, 37, 36]

The Reducer compares the values and produces:

1990 → 38

Therefore, the maximum temperature for that year is 38°C.

Q4. What challenges arise when processing large climate datasets?

Some major challenges include:

- Very large dataset size
- Data parsing and formatting issues
- Missing or invalid records
- Different temperature formats
- High processing requirements
- Storage requirements
- Network communication between nodes
- Efficient resource utilization
- Handling failures in distributed processing

MapReduce helps address these challenges by dividing the processing workload across multiple nodes.

Q5. How does Hadoop improve scalability and performance for weather data analytics?

Hadoop distributes large datasets across multiple machines and processes them in parallel. MapReduce divides the computation into smaller tasks that can run simultaneously.

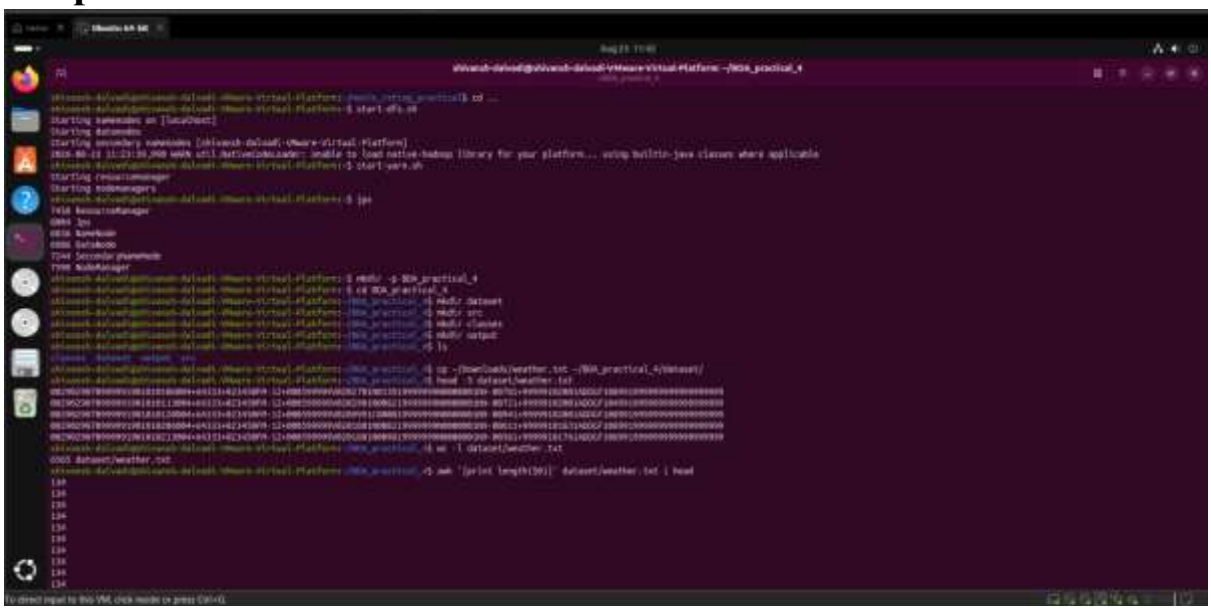
This provides:

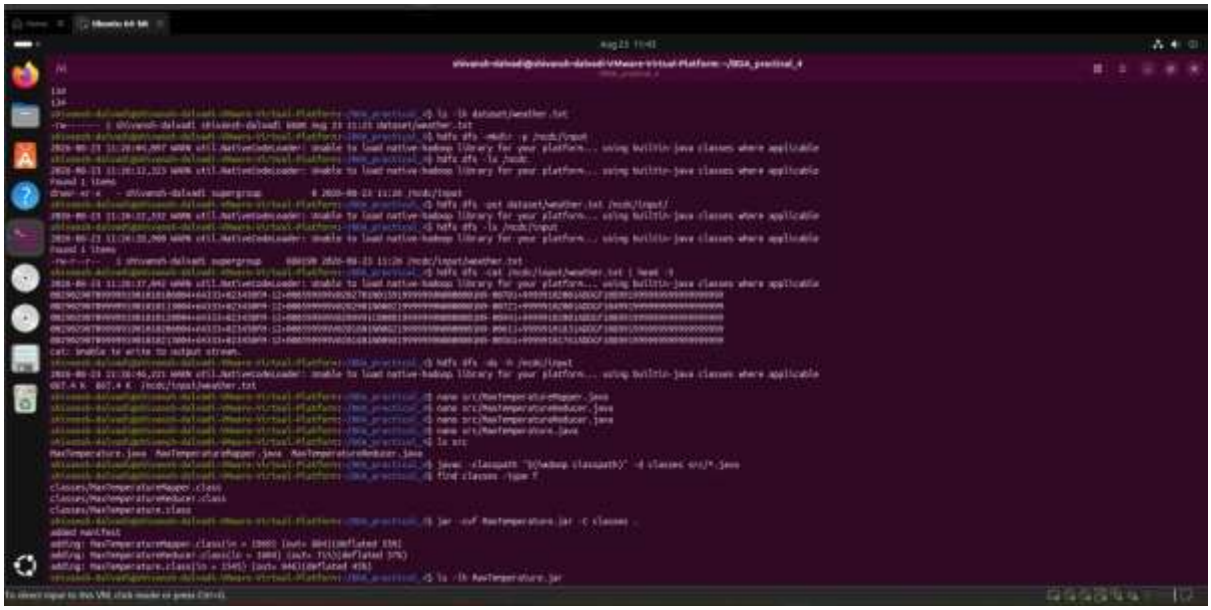
1. Scalability – additional machines can be added when data volume increases.
2. Parallel processing – multiple records can be processed simultaneously.
3. Fault tolerance – failed tasks can be re-executed.
4. High throughput – large datasets can be processed efficiently.
5. Distributed storage – large datasets can be stored across multiple nodes.

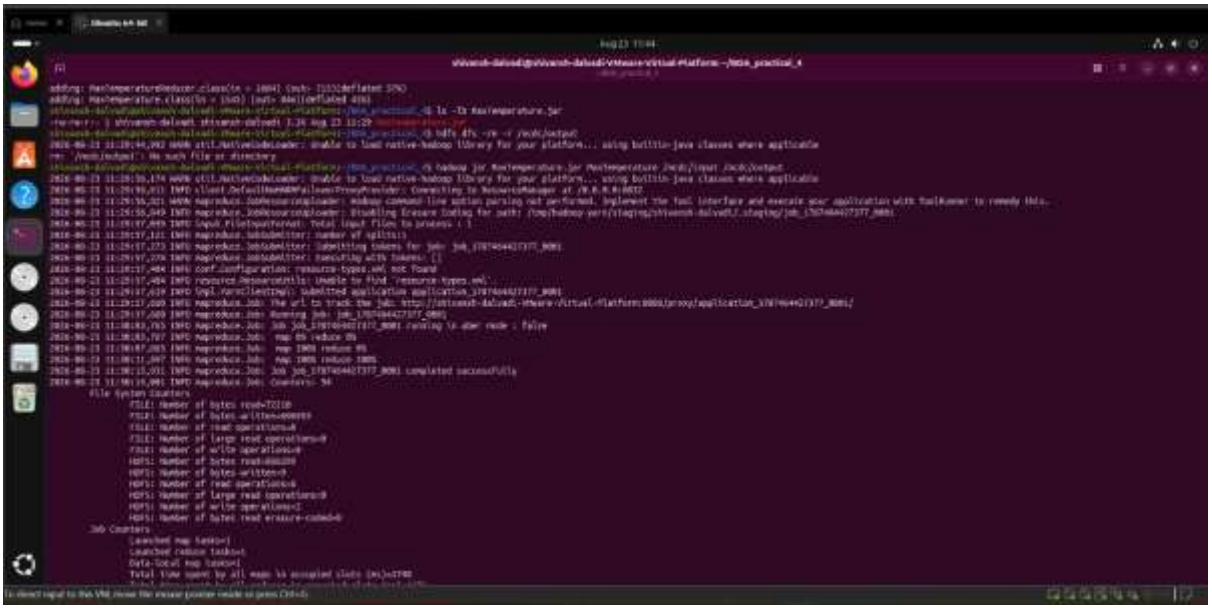
Supplementary Problems

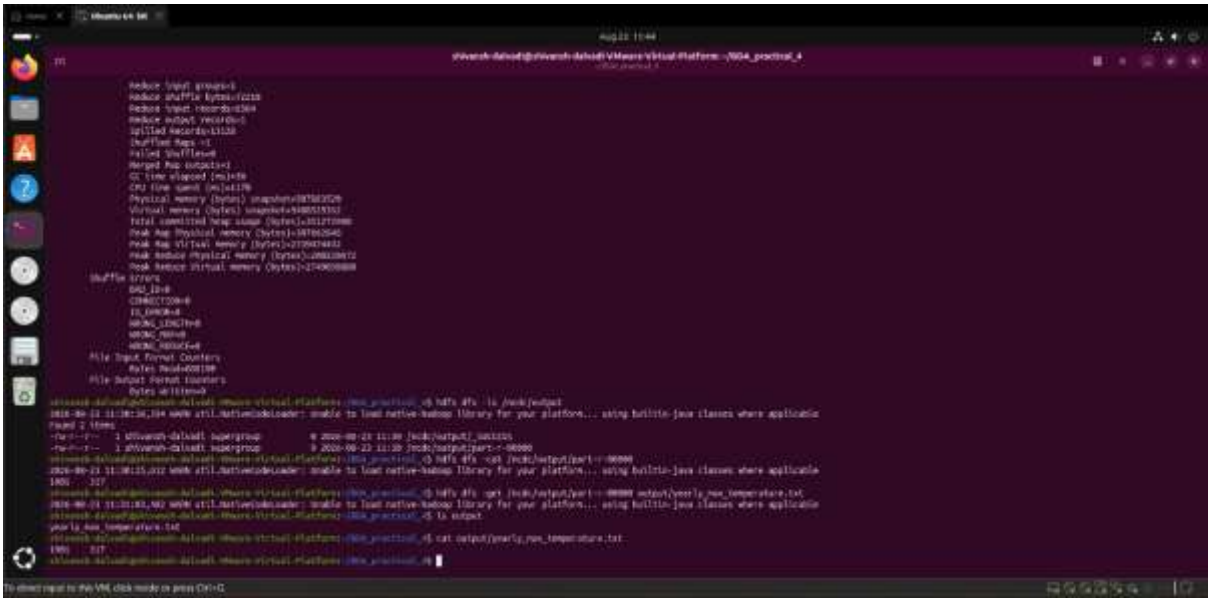
1. Find the minimum temperature recorded for each year.
2. Calculate the average yearly temperature.
3. Identify the year with the highest recorded temperature overall.

Output









Conclusion

The NCDC Weather Data Analysis using MapReduce practical was successfully completed. The NCDC weather dataset was uploaded into HDFS and processed using a Hadoop MapReduce program.

The Mapper extracted the year and temperature from each weather record and generated intermediate key-value pairs. The Reducer grouped the temperature values according to their respective years and determined the maximum temperature recorded for each year.

The practical provided a better understanding of MapReduce programming, data parsing, HDFS data management, distributed processing, and large-scale climate data analysis. The generated yearly maximum temperature results can further be used to study historical temperature trends and identify the hottest year in the dataset.